

STUDY PROTOCOL

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Bridging Respiratory Exposures, Asthma, and Environmental Health in Corpus Christi (BREATHE-CC): a prospective cohort study protocol

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Abstract

Background Childhood asthma disproportionately affects Hispanic children in the United States. Nevertheless, few longitudinal cohorts exist in Gulf Coast communities where environmental exposures may compound sociodemographic vulnerabilities. Corpus Christi, a majority-Hispanic city with high levels of petrochemical industrial operations, reports higher asthma prevalence than state and national averages. BREATHE-CC (Bridging Respiratory Exposures, Asthma, and Environmental Health in Corpus Christi) is a prospective cohort designed to investigate the role of air pollutants, household conditions, and social determinants of health in asthma exacerbations and wheezing phenotypes.

Methods Approximately 200 children less than 10 years of age with asthma will be enrolled from Driscoll Children's Hospital. Enrolled participants complete baseline assessments and are followed monthly for up to 18 months using a parent-reported modified ISAAC (International Study of Asthma and Allergies in Childhood) questionnaire that captures household exposures, asthma exacerbations, and wheezing episodes. Questionnaire responses will be validated and supplemented with electronic health record (EHR) reviews. Daily air pollutant concentrations (PM_{2.5}, PM₁₀, O₃, SO₂) from the Texas Commission on Environmental Quality (TCEQ) and U.S. Environmental Protection Agency (EPA) monitoring stations will be linked to participants' addresses. Generalized Additive Models (GAMs) and Poisson regressions will assess the associations between air pollutant exposures, household risk factors, asthma exacerbations, and wheezing phenotypes, adjusting for age, sex, body mass index (BMI), and other sociodemographic indicators. Group-Based Trajectory Modeling (GBTM) will identify latent phenotypes of wheezing.

Discussion This is a longitudinal cohort examining Hispanic children residing in the Gulf Coast petrochemical corridor. In particular, the study integrates granular geospatial environmental monitoring to link daily air pollutant

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exposure levels with asthma exacerbation risk. GBTM will also be used to identify latent wheezing phenotypes, especially focusing on differences between infectious and non-infectious wheezing. Recruitment will occur at a single hospital in the Coastal Bend region with a modest sample size, which may limit generalizability.

Registration Clinical trial number: not applicable. Open Science Framework [https://osf.io/tbg6y/overview?view_only=87309518c35b4d78a7bbe79d5fa70ceb].

Keywords Air pollutants, Asthma exacerbation, Children, Household, Wheezing

Introduction

Asthma is a chronic respiratory condition requiring continued management through specialist care and pharmacologic treatment. It typically presents a triad of symptoms, including cough, shortness of breath, and wheezing [1]. The prevalence and severity of childhood asthma vary by demographic, social, and environmental factors, underscoring the need to understand how these determinants contribute to asthma and respiratory symptom trajectories [2–4].

Wheezing, a primary symptom of asthma, commonly occurs during acute asthma exacerbations, often resulting in urgent care, hospitalization, and administration of corticosteroids for asthma management [5]. Children exhibit distinct wheezing phenotypes across early life, regardless of asthma diagnosis [6]. Associations between childhood wheezing trajectories and later asthma development depend on the mode, persistence, and timing of wheezing events, highlighting critical windows for environmental and social influences on asthma risk [7].

Corpus Christi, a coastal city in South Texas, experiences a semi-arid climate and is characterized by unique meteorological factors, complex environmental exposures, and sociodemographic disparities [8]. With an estimated population of 317,317, Corpus Christi is home to a growing economy and serves as a regional center for large-scale fabrication, marine research, maritime shipping, petrochemical manufacturing, and tourism [9, 10]. These industries contribute to environmental pollution, including air pollutants, such as ozone (O₃) and fine particulate matter (PM_{2.5}), prompting air quality and meteorological monitoring.

These environmental exposures, combined with social and housing conditions, may explain the high burden of childhood asthma in Corpus Christi, Texas, where prevalence is estimated at 7.8%, exceeding state (7.0%) and national averages (6.5%) [11–13]. Hispanic or Latino individuals constitute the largest demographic group (62.0%), while 17.1% of residents live in poverty, 20.8% lack health insurance, and 32.1% speak a language other than English [10]. This intersection of environmental conditions, sociodemographic vulnerability, and city infrastructure makes Corpus Christi an ideal setting to examine the impact of air pollutant exposures, household

living conditions, and social determinants of health on childhood asthma and respiratory symptoms.

Few cohorts focus on identifying risk and protective factors for chronic disease among Hispanic/Latino populations in the United States. The Hispanic Community Health Study/Study of Latinos (HCHS/SOL), a multicenter epidemiological study, is the largest Hispanic/Latino cohort to date, with the aim of understanding the role of environmental, genetic, and lifestyle factors in adult cardiovascular and respiratory health outcomes. Additionally, the Hispanic Community Children's Health Study/Study of Latino Youth, an ancillary study to HCHS/SOL, investigates the influence of parental and familial factors on youths' cardiometabolic risk profile [14]. These community-based cohorts span across four major metropolitan areas in the U.S., including Miami, Chicago, New York City, and San Diego.

BREATHE-CC (Bridging Respiratory Exposures, Asthma, and Environmental Health – Corpus Christi) is one of the first cohort studies in an industrial coastal city that examines the impact of environmental factors in early life on children's respiratory health among a predominantly Hispanic population. BREATHE-CC is uniquely positioned to longitudinally investigate how air pollution and housing conditions are associated with asthma exacerbations and wheezing trajectories in a high-risk population within a geographically underserved region. Findings from this cohort will inform targeted interventions to improve pediatric respiratory health outcomes, including alignment with national efforts such as the Environmental Influences on Child Health Outcomes (ECHO) program [15].

The primary research objectives of BREATHE-CC are to:

1. Assess the impact of geospatial and environmental factors, including residential proximity to industrial zones and ambient air pollutant levels of ozone and fine particulate matter, on childhood asthma exacerbation risk.
2. Identify potential household risk factors, including multigenerational living, cooking surface types, and secondhand smoke exposure, contributing to asthma exacerbation and wheezing events.

Methods

BREATHE-CC is a prospective cohort study at Driscoll Children's Hospital between 2025 and 2027. This study is a collaborative effort involving Texas A&M University – Corpus Christi, Driscoll Children's Hospital, and the Global Institute for Hispanic Health. It aims to investigate how air pollutant exposures, household living conditions, and broader social determinants of health influence childhood asthma and respiratory symptoms.

The study population includes children receiving asthma care at Driscoll Children's Hospital residing in the Coastal Bend Region, including Aransas, Bee, Brooks, Duval, Jim Wells, Kenedy, Kleberg, Live Oak, Nueces, Refugio, and San Patricio counties (Fig. 1). Participants are evaluated at baseline and followed monthly for up to 18 months via self-reported surveys. The BREATHE-CC cohort is longitudinally following asthmatic children, generating a comprehensive profile of environmental

exposures, household conditions, and respiratory health outcomes. As of June 2026, 50 participants have been enrolled in the BREATHE-CC cohort. Recruitment is actively ongoing with a target enrollment of 200 children by July 2027.

Inclusion and exclusion criteria

The study includes children less than 10 years of age receiving care at Driscoll Children's Hospital. Children must be diagnosed with asthma and reside in the Coastal Bend Region to be eligible to participate. Children with cystic fibrosis, congenital heart disease, primary immunodeficiencies, neuromuscular disorders impacting respiration, and non-asthma chronic lung diseases will be excluded from the study. All patient-facing materials, including recruitment and consent documents and survey questionnaires, are available in both English and Spanish to ensure cultural sensitivity and health literacy.

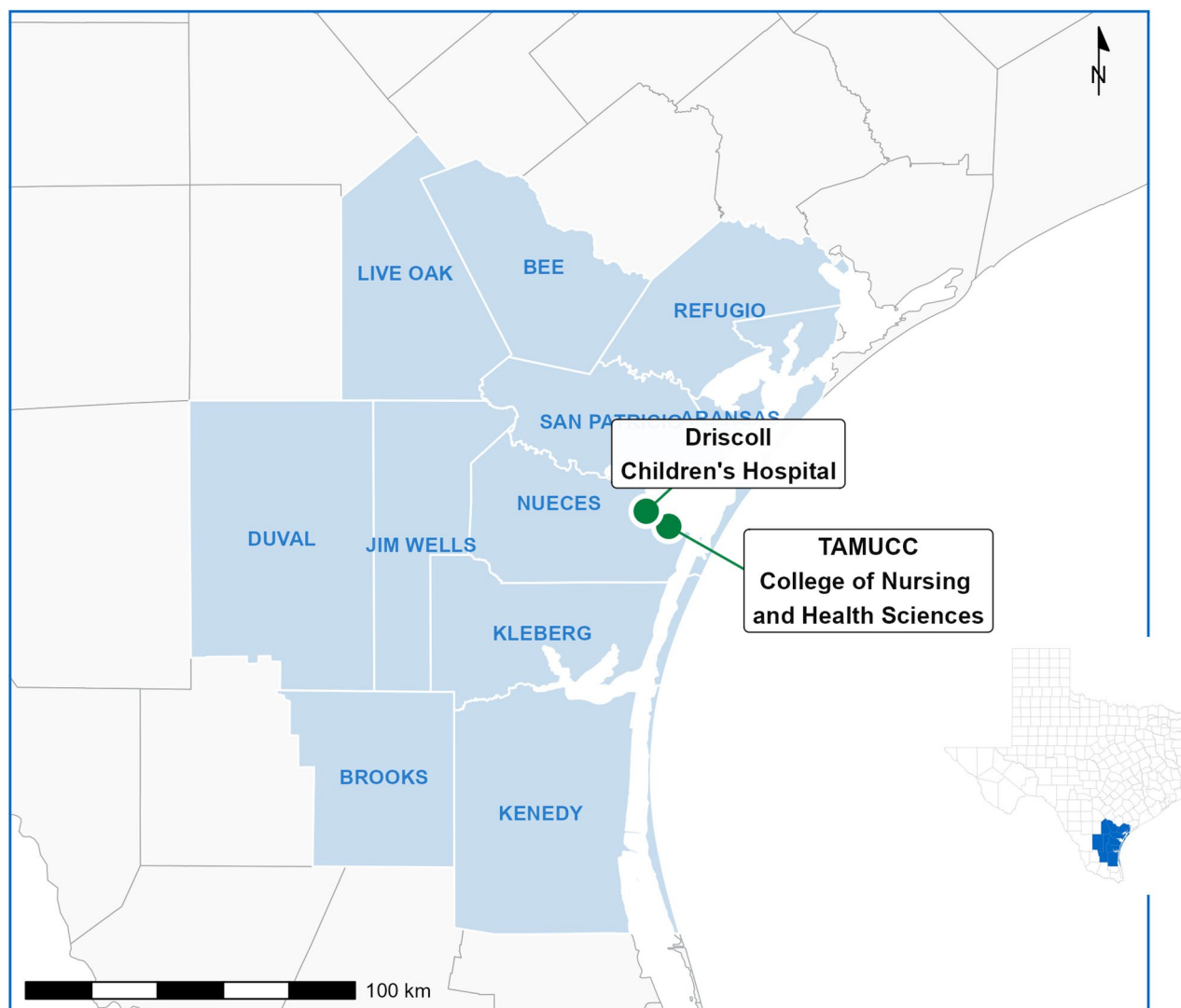


Fig. 1 Geographic Setting of the Study Population in the Coastal Bend Region, Texas, USA

Eligible participants receive gift cards upon completing study activities to maintain participant engagement throughout the follow-up period.

Recruitment and consent

Pediatric patients diagnosed with or being followed for asthma are identified through the pulmonology clinic, where patients are routinely seen for asthma management. Providers in the clinic introduce the study to eligible families and provide a warm handoff to trained Driscoll staff, who explain the study in detail, obtain informed consent, and complete the enrollment process. At this time, the baseline survey will be offered, or a link to complete the baseline survey will be provided.

Recruitment materials, including flyers and brochures, are available in the clinic setting to supplement provider referrals and help families understand the study. Study staff confirm eligibility through chart review to ensure that only patients meeting inclusion criteria are approached.

This clinic-centered approach, supported by provider-led handoffs and staff verification of eligibility, ensures that recruitment is both efficient and integrated into the care process for children with asthma. Figure 2 illustrates the workflow for recruitment and consent procedures.

Study protocol

Data sources

This project includes various data sources to address the impact of geospatial, environmental, and household factors on asthma exacerbation incidence. The main data sources include: (1) a modified ISAAC (International Study of Asthma and Allergies in Childhood) questionnaire survey; (2) electronic health records (EHRs) storing patient medical information; and (3) ambient air pollution data from local monitoring stations operated by the Texas Commission on Environmental Quality (TCEQ) and U.S. Environmental Protection Agency (EPA). To account for atmospheric conditions, meteorological and solar irradiance data will be sourced from OpenWeather-Map (OWM).

Outcomes

The primary outcomes of the study are asthma exacerbations and wheezing phenotypes. Asthma exacerbations will be obtained through self-reported information in follow-up surveys and via EHR review. Exacerbations are defined as unscheduled physician visits for asthma symptoms, emergency department visits or hospitalizations for asthma, use of systemic corticosteroids or rescue medications for worsening symptoms, or self-reported “asthma attack.”

Wheezing will be assessed using follow-up surveys to determine the presence of any wheeze, infectious wheeze,

and non-infectious wheezing [7]. Asthma medication management will also be assessed through questionnaires and validated via EHRs to determine adherence to prescriptions and asthma or wheeze control efficacy.

Geospatial and environmental exposures

Daily ambient concentrations of air pollutants ($PM_{2.5}$, PM_{10} , O_3 , SO_2) measured by the TCEQ and EPA will be collected from monitoring stations at Corpus Christi West, Corpus Christi Tuloso, Corpus Christi Huisache, and Dona Park. Air pollutant exposures will be used as covariates in regression models. Participants' residential addresses will be obtained from REDCap (Research Electronic Data Capture) enrollment surveys and geocoded to geographic coordinates for spatial linkage and utilization in modeling. For participants with complex housing situations, questionnaires will ascertain the amount of time spent at each residence with exposure estimates adjusted commensurately.

Meteorological and solar irradiance data

To pair atmospheric conditions with air pollutant measurements, high-resolution meteorological and solar irradiance data will be obtained from OWM. These parameters include temperature ($^{\circ}C$), relative humidity (%), wind speed (m/s), wind direction ($^{\circ}$), atmospheric pressure (hPa), precipitation (mm), as well as solar irradiance components (W/m^2), including Global Horizontal Irradiance (GHI), Direct Normal Irradiance (DNI), and Diffuse Horizontal Irradiance (DHI). These parameters are essential for characterizing conditions that influence petrochemical reactions, particulate formation, air pollutant dispersion, and daily and seasonal variability in air pollutant levels.

Assessments

Upon enrollment, a baseline assessment is administered to the parent or guardian in the clinic to collect information on key sociodemographic characteristics, family history of atopic disease, birth history, and household environment (Supplement 1). Subsequently, parents or guardians are asked to complete a brief, monthly survey via REDCap to document the occurrence of asthma exacerbation, wheezing episodes, medication use, and hospital admissions (Supplement 2). Each monthly survey records changes in residence, which prompts a separate housing survey (Supplement 3). Primary and secondary household safety perceptions will be collected quarterly. Participant EHRs will also be reviewed quarterly to extract clinical data, including laboratory test results, diagnoses, medications, allergies, and clinical notes.

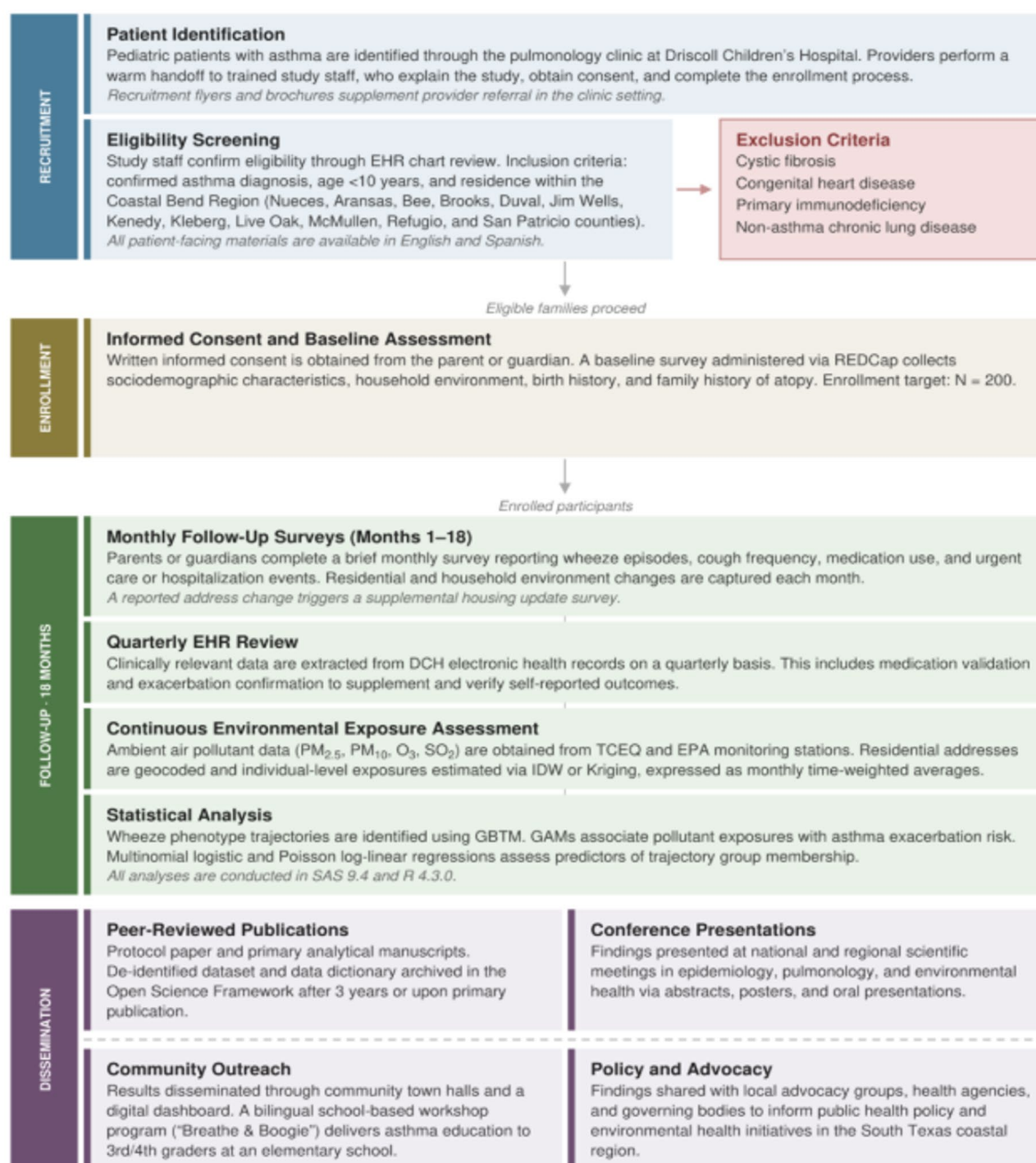


Fig. 2 BREATHE-CC Study Workflow: Recruitment, Enrollment, Follow-Up, and Dissemination

Sample size

A minimum of 100–150 subjects is generally required for stable estimation of 3–6 trajectory groups, specifically when the smallest group contains at least 5% of the total sample [16]. Our expected follow-up structure (6–18

repeated binary wheezing measures per participant) will allow us to construct a longitudinal profile of wheezing suitable for Group-Based Trajectory Modeling (GBTM).

We conducted a power analysis using a two-sample t-test framework to detect differences in wheezing event

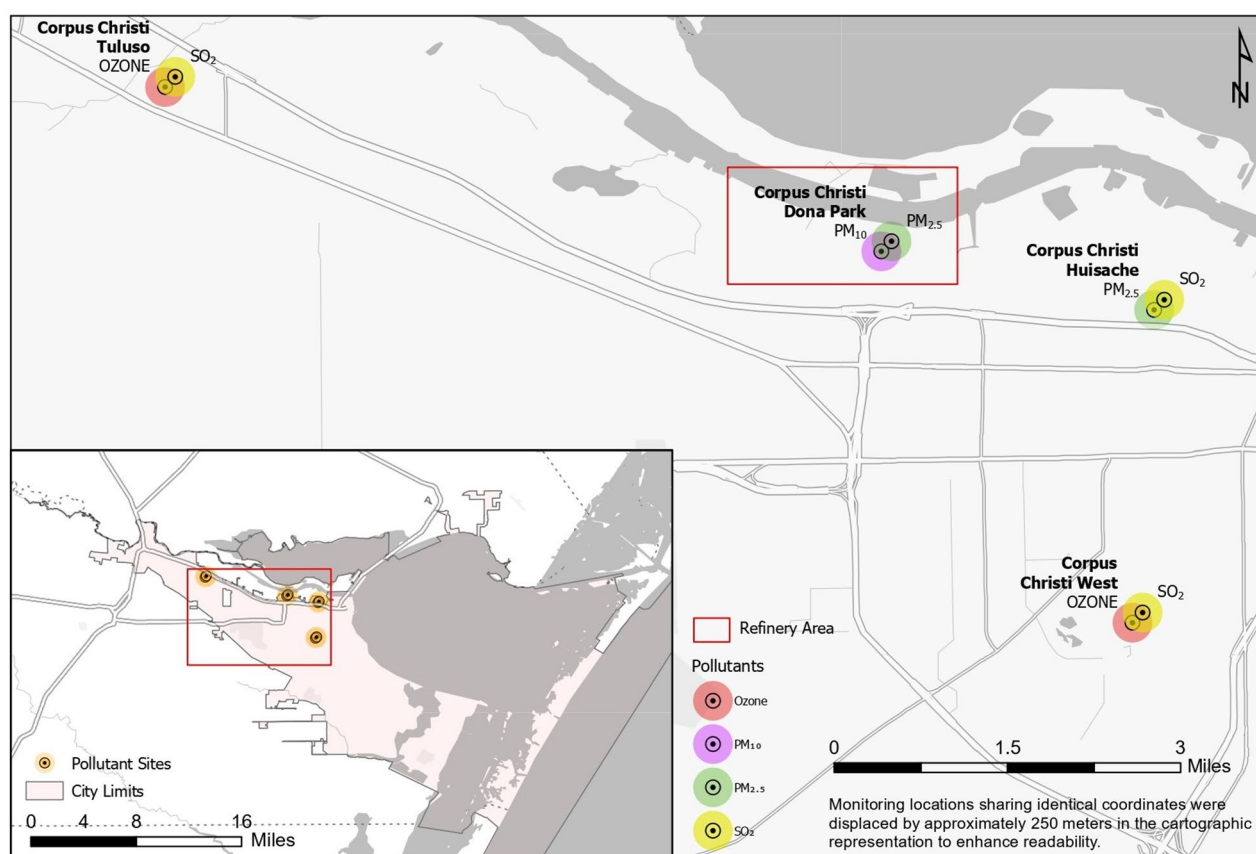


Fig. 3 Air Quality Monitoring Locations in Corpus Christi, Texas

counts between two groups over a 12-month follow-up period. This analysis evaluated a range of expected mean differences in wheeze counts (2, 4, 6, and 8 events) and two plausible levels of variability (standard deviations of 2.5 and 5) [7]. Based on these simulations, approximately 150 participants are needed to discern differences when variability is high, though that number decreases to as few as 75 participants with low variability.

Simulated Poisson regressions with 12-month wheeze count data were used to estimate statistical power to detect Incidence Rate Ratios (IRRs) of 1.3, 1.5, and 2.0. For IRRs greater than 1.5, a sample size of 150 participants achieves at least 80% power, even accounting for overdispersion. While a smaller IRR would require a larger sample size, the proposed sample size is sufficient to identify moderate environmental associations consistent with acute exposure effects. Based on these power calculations, we aim to enroll at least 200 participants with an analytic sample size closer to 150 based on attrition.

Statistical analysis

The statistical analysis of the cohort will proceed in three stages: (1) geospatial context and exposure measurement of air pollutant concentrations, (2) generation of

wheezing and asthma exacerbation trajectories, and (3) evaluation of associations between air pollutant exposures and household conditions with wheezing phenotypes and asthma outcomes. All models will account for repeated observations within participants. Statistical analyses will be conducted in SAS 9.4 (SAS Institute, Cary, NC) and R version 4.3.0 (R Foundation for Statistical Computing, Vienna, Austria).

Daily concentrations of air pollutants ($PM_{2.5}$, PM_{10} , O_3 , SO_2) will be extracted from EPA and TCEQ monitoring stations across the sample area (Fig. 3).

Patient residential addresses and monitoring stations will be geocoded. Generalized Additive Models (GAMs) will examine associations between air pollutant exposure and the incidence of asthma exacerbation, adjusting for meteorological influences. GAMs are widely used to evaluate the impact of exposure-outcome relationships, such as air pollution and respiratory disease, and can enhance predictive capacity by incorporating spatial and temporal components, including geographical location and time of day [17–19].

Wheezing incidence will be determined through monthly follow-up surveys. Three wheezing types will be identified through the modified ISAAC questionnaires: any wheeze, infectious wheeze, and non-infectious

wheeze [7]. GBTM will identify trajectories for each wheezing phenotype using Proc TRAJ [20]. We will fit models with 2 to 5 trajectories using the AutoTRAJ Macro, selecting the optimal number based on the Bayesian Information Criterion (BIC), average posterior probability (> 0.70), and clinical interpretability [21]. Missing data will be handled via the maximum likelihood method.

Clinical outcome data will be extracted from questionnaires and EHRs. The primary outcomes of interest are asthma exacerbations (unscheduled clinic or hospital visits for asthma, reported asthma attacks, and use of asthma medications). Multinomial logistic regression will identify risk factors for trajectories with higher wheeze frequency and asthma. Predictors will include environmental exposures (air pollutant exposure), household characteristics (multigenerational living, cooking surfaces, smoking), and birth history (birth order, birth weight, family history of atopy) while adjusting for age, sex, body mass index (BMI), and other sociodemographic indicators. Log-linear regressions with a Poisson distribution will be used to compare asthma exacerbation rates across trajectories.

Discussion

Significance

Pediatric asthma exacerbations in children remain a leading cause of emergency department visits and hospitalizations in the United States. Wheezing is a key symptom of asthma that manifests as phenotypic trajectories, which can differentially predict asthma diagnosis and severity. Despite these well-documented outcomes, few prospective cohorts examine the intersection of environmental exposures with respiratory outcomes in Hispanic children. BREATHE-CC addresses this gap through a longitudinal analysis of asthma exacerbations and wheeze phenotypes in asthmatic children residing in the Corpus Christi region. We further integrate ambient air pollution, household factors, and EHR-validated clinical records to better characterize the exposome associated with asthma morbidity in this vulnerable population.

Strengths and limitations

BREATHE-CC is geographically situated in the Gulf Coast petrochemical corridor, an underrepresented region in environmental epidemiology. Therefore, BREATHE-CC is well-positioned for geospatial exposure assessment due to the presence of air pollutant monitoring stations in the region. Another strength of the study is the use of GBTM, which allows for the transformation of discrete incidence of wheeze into clinically meaningful phenotypes with distinct risk profiles.

Several limitations warrant consideration. The target sample size of 200 is adequate for trajectory modeling and detection of moderate effect sizes through Poisson

regression. However, smaller subgroup analyses may be limited to descriptive statistics due to power restrictions. Selection bias is another limitation since the study population includes pediatric asthma patients rather than members of the general population, meaning observed exposure-outcome associations may not fully represent the general population. Moreover, the study utilizes parent-reported surveys, which may introduce recall bias. Quarterly EHR reviews will supplement these questionnaires to validate their responses.

Community collaboration

The public was not directly involved in the design of this study protocol. However, research priorities were informed by health disparities identified by key community partners, including Texas A&M University – Corpus Christi, Driscoll Children's Hospital, and the Global Institute for Hispanic Health.

Dissemination

Patients and the public will be engaged in the dissemination of study findings. Results will be shared with local advocacy groups and government agencies to inform public health policies and interventions. Findings will also be disseminated through peer-reviewed publications and presentations at applicable national conferences.

At the community level, results and educational materials will be delivered through workshops and events for teachers, students, and families, as well as through a digital dashboard. As a community extension of BREATHE-CC, a bilingual outreach program has been developed to promote early asthma awareness among elementary schoolchildren through educational sessions that focus on identifying asthma triggers, practicing breathing techniques, and understanding basic respiratory health concepts. The program is to be implemented across two settings, which are elementary school-based sessions and public library workshops open to the community, with a dual setting model intended for replication across additional schools and library branches.

Abbreviations

BIC	Bayesian Information Criterion
BMI	Body mass index
BREATHE-CC	Bridging Respiratory Exposures, Asthma, and Environmental Health in Corpus Christi
DCHIRB	Driscoll Children's Hospital Institutional Review Board
DHI	Diffuse Horizontal Irradiance
DNI	Direct Normal Irradiance
ECHO	Environmental Influences on Child Health Outcomes
EHR	Electronic health record
EPA	Environmental Protection Agency
GAMs	Generalized Additive Models
GBTM	Group-Based Trajectory Modeling
GHI	Global Horizontal Irradiance
HCHS/SOL	Hispanic Community Health Study/Study of Latinos
IRR	Incidence Rate Ratio
ISAAC	International Study of Asthma and Allergies in Childhood

OWM	OpenWeatherMap
O ₃	Ozone
PM _{2.5}	Fine particulate matter
PM ₁₀	Coarse particulate matter
REDCap	Research Electronic Data Capture
SO ₂	Sulfur dioxide
TCEQ	Texas Commission on Environmental Quality

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12889-026-28265-5>.

Supplementary Material 1.

Supplementary Material 2.

Supplementary Material 3.

Authors' contributions

DEW: Conceptualization, Methodology, Formal Analysis, Writing – Original Draft, Writing – Review & Editing. AM: Data Curation, Project Administration, Writing – Review & Editing. HA: Investigation, Resources, Writing – Review & Editing. MK: Data Curation, Visualization, Writing – Review & Editing. LJ: Methodology, Software, Writing – Review & Editing. ER: Investigation, Resources. MA-H: Investigation, Resources. NJ: Methodology, Writing – Review & Editing. JR: Supervision, Investigation, Resources. RM: Conceptualization, Methodology, Funding Acquisition, Supervision, Project Administration, Writing – Original Draft, Writing – Review & Editing.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

This study adheres to the Declaration of Helsinki. Ethical approval was granted by the DCHIRB (FY2025-85). Written informed consent is obtained from parents/guardians. Study data are collected and managed using REDCap hosted at Texas A&M University – Corpus Christi. REDCap provides audit trails for tracking data changes and user activity. Furthermore, a data use and management protocol has been developed to secure data access and protect personal identifying information. REDCap study IDs are unique to patients, and the linkage between BREATHE-CC ID and DCH ID will be stored on a separate server accessible only to key study personnel. Patient addresses and date of birth will be anonymized by authorized personnel and transformed into calculated variables (age and geocoded address with a jitter).

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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